Data Science & Artificial Intelligence

Machine Learning

Bayesian Learning

DPP 01

Discussion Notes

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[NAT]

#Q. Consider a naive Bayes classifier with 3 boolean input variables, X_1, X_2 and X_3, and one boolean output, Y.

How many parameters must be estimated to train such a naive Bayes classifier? (you need not list them unless you wish to, just give the total)

Sol 3 dimensions $x_1, x_2, x_3 \rightarrow$ Boolean binary 0, 1

Class $Y \Rightarrow$ Boolean 0, 1

$\underbrace{P_{y_1} \; P_{y_2}}_{\text{Class Probab}} \rightarrow 1$

$2 + 3 \times 2 \times 2 \Rightarrow \underline{14} \rightarrow 7$

	Y=0	Y=1
$X_1=0$	✓	✓
$X_1=1$	✓	✓
$X_2=0$	✓	✓
$X_2=1$	✓	✓
$X_3=0$	✓	✓
$X_3=1$	✓	✓

[MCQ]

#Q. Match the items in Column 1 with the items in Column 2 in the following table:

Column 1	Column 2
(p) Principal Component Analysis	(i) Discriminative Model
(q) Naïve Bayes Classification	(ii) Dimensionality Reduction
(r) Logistic Regression	(iii) Generative Model

A (p) (iii), (q) - (i), (r) - (ii)

B (p) (ii), (q) - (i), (r) - (iii)

C (p) (ii), (q) - (iii), (r) - (i)

D (p) (iii), (q) - (ii), (r) - (i)

[MCQ]

#Q. Suppose you have a three class problem where class label $y \in 0,1,2$ and each training example X has 3 binary attributes $X_1, X_2, X_3 \in 0,1$. How many parameters do you need to know to classify an example using the Naive Bayes classifier?

Sol $3 + 3 \times 2 \times 3 \Rightarrow \underline{\underline{21}}$

A 5

B 9

C 21

D 13

E 23

[MCQ]

#Q. Consider the two class problem where class label $y \in \{T, F\}$ and each training example X has 2 binary attributes $X_1, X_2 \in \{T, F\}$.

How many parameters will you need to know/estimate if you are to classify an example using the Naivs Bayes classifier?

Sol $2 + 2 \times 2 \times 2 \Rightarrow 10 \rightarrow 5$

A 5

B 8

C 3

D 7

[NAT]

#Q. Suppose that you are trying to solve a binary classification problem, and your data set has 4 at-tributes. Each attribute can take 3 possible values.

If you modeled the full joint distribution of the attributes and the class label, how many parameters would you need?

Sol

2 Class → 1

4 dimension → 3 possible Values

$2 + 4 \times 3 \times 2 \Rightarrow \underline{26} \longrightarrow \boxed{1 + 4 \times 3 \times 2 \rightarrow 25}$

[NAT]

#Q. Consider a classification problem with two classes and n binary attributes.
How many would you need to learn with a Naive Bayes classifier?
How many parameters would you need to learn with a Bayes optimal classifier?

(Not in Syllabus)

$2 + n \times 2 \times 2 \Rightarrow \boxed{(4n+2)}$

$1 + (n \times 2 \times 1) \Rightarrow (1+2n)$

[MCQ]

#Q. Consider a classification problem with two binary features, $x_1, x_2 \in \{0,1\}$. Suppose $P(Y = y) = 1/32$, $P(x_1 = 1 \ Y = y) = y/46$, $P(x_2 = 1 \mid Y = y) = y/62$. Which class will naive Bayes classifier produce on a test item with $x_1 = 1$ and $x_2 = 0$?

A 16

B 26

C 31 ✓

D 32

Sol $P(y)\, P(x_1 = 1/y)\, P(x_2 = 0/y)$ max

$$\frac{1}{32} \times \frac{y}{46}\left(1 - y/62\right)$$

$$\frac{d}{dy} = 0 \Rightarrow \frac{1}{32}\left[\frac{1}{46} - \frac{2y}{46 \times 62}\right] = 0$$

$$y = 31$$

[MSQ]

#Q. What is the total number of parameters needed to model $P(y, x_1, x_2, x_3, \ldots, x_d)$?

- k is the number of possible values Y can take
- t is the number of value of each x_i where $i \in \{1, 2, \ldots, d\}$.

A $k \cdot t^d - 1$

B $k \cdot t^{d-1}$

C $k \cdot (t^d - 1) + (k - 1)$

D ~~$t^d \cdot k$~~ (kdt)

- No of Parameter $k + k \times d \times t$

Class Probab

$\rightarrow (k + kdt)$

we can neglect $k \Rightarrow kdt$

[NAT]

#Q.

A_1	A_2	Class Label Y
True	True	+
True	True	+
True	False	–
False	False	+
False	True	–
False	False	–
False	False	–
True	False	+

$$P(y=+/T,T) = \frac{P(T,T/+) \cdot P_+}{P(T,T)}$$

$$P(y|x) = \frac{P(x/y)\, P_y}{P_x}$$

What is the probability $P(Y = + \mid A_1 = \text{True}, A_2 = \text{True})$ by applying the Naïve Bayes classifier (show each step in computing the probability)? do What label should you predict for input (A_1 = True, A_2 = True) based on your result above ?

[NAT]

#Q.

A_1	A_2	Class Label Y
True	True	+ ✓
True	True	+ ✓
True	False	–
False	False	+ ✓
False	True	–
False	False	–
False	False	–
True	False	+ ✓

$$P(y=+/T,T) = \frac{P(T,T/+) \cdot P_+}{P(T,T)}$$

$$P(y=-/T,T) = \frac{P(T,T/-)\, P_-}{P(T,T)}$$

$$\left(P_+ = \frac{4}{8} = \frac{1}{2},\ P_- = \frac{4}{8} = \frac{1}{2}\right)$$

What is the probability $P(Y = + \mid A_1 = \text{True}, A_2 = \text{True})$ by applying the Naïve Bayes classifier (show each step in computing the probability)? do What label should you predict for input (A_1 = True, A_2 = True) based on your result above ?

[NAT]

#Q.

A_1	A_2	Class Label Y
True	True	+ ✓
True	True	+ ✓
True	False	−
False	False	+ ✓
False	True	−
False	False	−
False	False	−
True	False	+ ✓

$$P(T,T/+) = P(T/+)\,P(T/+)$$
$$= \frac{3}{4} \times \frac{2}{4} \Rightarrow \frac{3}{8}$$

$$P(T,T/-) = P(T/-)\,P(T/-)$$
$$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

What is the probability $P(Y = + \mid A_1 = \text{True}, A_2 = \text{True})$ by applying the Naïve Bayes classifier (show each step in computing the probability)? do What label should you predict for input (A_1 = True, A_2 = True) based on your result above ?

$$P(y=+/T,T) = \frac{\frac{3}{8} \times \frac{1}{2}}{P(T,T)}$$

$$P(y=-/T,T) = \frac{\frac{1}{16} \times \frac{1}{2}}{P(T,T)}$$

add = 1

$$\frac{3/16}{P(T,T)} + \frac{1/32}{P(T,T)} = 1$$

$$P(T,T) = .21875$$

$$\underbrace{P(y=+/T,T) = \frac{6}{7}}, \quad P(y=-/T,T) = \frac{1}{7}$$

So naive Bayes Predict + Class

[NAT]

#Q. Suppose you have the following training set with three boolean input x, y and z, and a boolean output U.

x	y	z	U
1	0	0	0
0	1	1	0
0	0	1	0
1	0	0	1
0	0	1	1
0	1	0	1
1	1	0	1

Sol $P_{u=0} = 3/7 \quad P_{u=1} = 4/7$

$P(u=0/x=0, y=1, z=0) =$

$$\frac{P(x=0, y=1, z=0/u=0)\, P_{u=0}}{P(x=0, y=1, z=0)}$$

$P(x=0/u=0)\, P(y=1/u=0)\, P(z=0/u=0)$

$2/3 \times \frac{1}{3} \times \frac{1}{3}$

Suppose you have to predict U using a Naivs Bayes classifier. After learning is complete what would be the predicted probability $P(U = 0 | x, = 0 y = 1, z = 0,)$?

[NAT]

#Q. Suppose you have the following training set with three boolean input x, y and z, and a boolean output U.

x	y	z	U
1	0	0	0
0	1	1	0
0	0	1	0
1	0	0	1
0	0	1	1
0	1	0	1
1	1	0	1

Sol $P_{u=0} = 3/7 \quad P_{u=1} = 4/7$

$P(u=1/x=0, y=1, z=0)$

$$= \frac{P(x=0, y=1, z=0/u=1)\, P(u=1)}{P(x=0, y=1, z=0)}$$

$P(x=0/u=1)\ P(y=1/u=1)\ P(z=0/u=1)$

$2/4 \times 2/4 \times 3/4$

Suppose you have to predict U using a Naivs Bayes classifier. After learning is complete what would be the predicted probability $P(U = 0 | x' = 0 y = 1, z = 0')$?

$$P(u=0 / x=0, y=1, z=0) = \frac{2/63}{P(x=0, y=1, z=0)} \longrightarrow 8/35$$

$$P(u=1, x=0, y=1, z=0) = \frac{3/28}{P(x=0, y=1, z=0)} \longrightarrow \frac{27}{35}$$

$$P(x=0, y=1, z=0) = 5/36$$

$\Rightarrow$ and we assign class 1 to the test point.

[NAT]

#Q. Suppose you have the following training set with three boolean input z, y and z, and a boolean output U.

x	y	z	U
1	0	0	0
0	1	1	0
0	0	1	0
1	0	0	1
0	0	1	1
0	1	0	1
1	1	0	1

Sol $P(u=0/x=0) = \dfrac{P(x=0/u=0)\, P_{u=0}}{P_{x=0}}$

$2/3$

$\dfrac{3}{7}$

$P(u=1/x=0) = \dfrac{P(x=0/u=1)\, P_{u=1}}{P_{x=0}}$

$2/4$

$4/7$

Suppose you have to predict U using a Naivs Bayes classifier. After learning is complete what would be the predicted probability $P(U = 0|x = 0)$?

$$P(u=0|x=0) = \frac{2/7}{P(x=0)}$$

$$P(u=1|x=0) = \frac{2/7}{P(x=0)}$$

$\rightarrow$ add $= 1$

$$P(x=0) = 4/7$$

$$\boxed{P(u=0|x=0) = .5}$$

[MCQ]

#Q. Again, consider the two class problem where class label $y \in \{T, F\}$ and each training example X has 2 binary attributes $x_1, x_2 \in \{T, F\}$. How many parameters will you need to estimate if we do not make the Naïve Bayes conditional independence assumption?

A 3

B 5

C 7

D 8

done. $2 + 2 \times 2 \times 2 = 10 \rightarrow 5$

[NAT]

#Q. Consider a Bayes classifier with 3 boolean input variables, x_1, x_2, and x_3 and one boolean output Y.

How many parameters must be estimated to train such that Bayes classifier?

Sol $2 + 3 \times 2 \times 2 \Rightarrow \underline{\underline{14}} \rightarrow 7$ ✓

[NAT]

2 Class

#Q. Suppose that you are trying to solve a binary classification problem, and your data set has 4 attributes. Each attribute can take 3 possible values If you modeled the ~~full joint distribution~~ of the attributes and the class label, how many parameters would you need? naive Bayes

Sol

$2 + 4 \times 2 \times 3 \Rightarrow 26$

25

[MCQ]

#Q. In a naive Bayes classifier, if the probability of class A given the predictor variables is 0.7 and the probability of class B given the same predictor variables is 0.3, what is the predicted class for a new data point?

A. Class A ✓

B. Class B

C. Cannot be determined

D. Depends on the feature space

$P(A/x) = .7$

$P(B|x) = .3$

Rule $\quad P(C_1/x) \underset{C_1 \quad C_2}{><} P(C_2/x)$

By Rule Clan A

[MCQ]

#Q. Consider a K-class dataset containing V points, where each sample is described by D continuous features. You decide to use a Gaussian Naive Bayes classifier. How many parameters need to be estimated for this model?

A. 2DK + K ✓

B. DK

C. 3DK

D. 2D + K

- Each Sample has D Cont. features
- K classes

Now we break data Class wise

$X^1/\text{class } 1$ → μ, σ

$X^1/\text{class } 2$ → μ, σ

--

$X^1/\text{class } K$ → μ, σ

Each dimension broken into K Class set and we find μ, σ for each set.

in 1 feature $\rightarrow$ we need $2 \times K$ parameter

in D features $\rightarrow$ " " $2 \times D \times K$ "

and we need K Class probabilities also

$\Rightarrow 2DK + K$

[MCQ]

#Q. Given the training set below with boolean input features x, y, z and a boolean output U, what would be the predicted probability $P(U = 0 \mid x = 0, y = 1, z = 0)$ after training a Naive Bayes classifier?

x	y	z	u
1	0	0	0
0	1	0	0
0	0	1	0
1	0	0	1
0	0	1	1
0	1	0	1
1	1	0	1

A. 6/35

B. 7/35

C. 8/35 ✓

D. 9/35

[MCQ]

#Q. Given a dataset with K binary-valued attributes (where K > 2) for a two-class classification task, the number of parameters to be estimated for learning a naïve Bayes classifier is

A. $2^k + 1$

B. $2K + 1$

C. $2^{(k+1)} + 1$

D. $K^2 + 1$

- Sol • K Binary attributes
  - 2 Classes → $P(y=1)$ → I can find $P_{y=0}$
  - No of parameters $= 2 + K \times 2 \times 2$
    $= \boxed{2 + 4k} \rightarrow 1 + 2k$
  - $P(x_1 = 0/y) \rightarrow P(x_1 = 1/y)$

THANK - YOU